



CLIENT DELIVERABLE

SAOS iOS Certificate Trust Plan

Systack Documentation

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ROLE	COLOR	HEX
Headers, CTAs	Navy	 #001a2d
Navy Light	Navy Light	 #002845
Secondary accents	Teal	 #007da9
Primary buttons, links	Cyan	 #00a1db
Gradients	Cyan Bright	 #00c5e0
Backgrounds	Gray 50	 #f8fafc
Cards	Gray 100	 #f1f5f9
Borders	Gray 200	 #e2e8f0
Muted text	Gray 400	 #94a3b8
Body text	Gray 600	 #475569
Headings	Gray 800	 #1e293b
Success	Green	 #22c55e
Error	Red	 #ef4444

ROLE	COLOR	HEX
Accent highlights	Purple	 #8b5cf6

SAOS iOS Safari `.ts.net` Certificate Trust — Resolution Plan

Date: 2026-06-30 **Status:**  IN PROGRESS — Plan created, awaiting implementation

Problem

iOS Safari blocks access to `*.ts.net` Tailscale URLs with certificate error: - **Error:** “This Connection Is Not Private” / “Certificate Invalid” - **Root cause:** Tailscale’s `.ts.net` certificates are trusted on macOS but NOT automatically trusted on iOS - **Impact:** Customers cannot access SAOS dashboard on iPhone/iPad via Tailscale

Root Cause Analysis

PLATFORM	TAILSCALE TRUST	MECHANISM
macOS	 Trusted	Tailscale system extension registers certificates with Keychain
iOS	 NOT Trusted	No system extension model on iOS; <code>.ts.net</code> certs from LetsEncrypt need manual trust
Android	 Varies	May work, may need manual trust depending on ROM

The `.ts.net` domains use LetsEncrypt certificates. iOS requires explicit user action to trust certificates for domains that aren’t preloaded in the HSTS list or part of a trusted app profile.

Solutions (Ranked by Feasibility)

Option 1: Custom Domain + Valid SSL (RECOMMENDED)

What: Point a custom domain (e.g., `dashboard.systack.net`) to the Tailscale IP via A record, use a real SSL certificate. **How:** 1. Buy/acquire `dashboard.systack.net` (or use existing `systack.net`) 2. Add A record pointing to the Tailscale IPv4 (100.x.x.x) or public IP 3. Use Tailscale serve with `--https` but override the cert 4. OR: Use Cloudflare Tunnel (cloudflared) instead of Tailscale serve — gives you a real public domain

Pros:  Works on all devices natively, professional appearance

Cons: Requires public domain, slightly more complex setup

Effort: 2 hours

Cost: \$0 (Cloudflare Tunnel free tier) or use existing domain

Option 2: Cloudflare Tunnel (Zero Trust)

What: Replace Tailscale serve with Cloudflare Tunnel — no `.ts.net` needed. **How:** 1. Install `cloudflared` on the MacBook Air 2. Create tunnel in Cloudflare dashboard 3. Point tunnel to `localhost:8768` 4. Assign hostname like `saos.systack.net` or `dashboard.systack.net` 5. Cloudflare handles SSL automatically

Pros: ✅ No cert issues on any device, built-in DDoS protection, analytics

Cons: Requires Cloudflare account, dependency on Cloudflare

Effort: 1 hour

Cost: \$0 (free tier covers this)

Option 3: Manual iOS Certificate Install

What: Have users manually install and trust the Tailscale/LetsEncrypt certificate. **How:** 1. Export the `.ts.net` certificate from the server 2. Host it at a trusted URL 3. User downloads the `.cer` file on iOS 4. User goes to Settings → General → VPN & Device Management → installs profile 5. User goes to Settings → About → Certificate Trust Settings → enables trust

Pros: Works with existing setup

Cons: ❌ Terrible UX, requires user action every cert renewal (90 days), friction kills adoption

Effort: 30 min to build docs, infinite user friction

Option 4: Tailscale App + MagicDNS (Partial)

What: Use the Tailscale iOS app — once connected, `.ts.net` names resolve. **How:** 1. Install Tailscale app on iOS 2. Connect to the tailnet 3. Access `phillips-macbook-air.tail573d57.ts.net/dashboard/`

Current state: ❌ Even with Tailscale app connected, Safari still shows cert error for `.ts.net`

Why: The cert is valid (LetsEncrypt) but iOS Safari doesn't auto-trust it without user interaction.

RECOMMENDED APPROACH: Option 2 (Cloudflare Tunnel)

Why This Option

- Free (Cloudflare Tunnel is \$0)
- Zero cert issues — Cloudflare issues and manages real certs
- No `.ts.net` dependency
- Works on any device, any browser
- Adds DDoS protection and analytics as bonus
- Can keep Tailscale for backend mesh (don't need to remove it)

Implementation Steps

STEP	ACTION	TIME
1	Verify Cloudflare account for <code>systack.net</code>	5 min

STEP	ACTION	TIME
2	Install <code>cloudflared</code> on MacBook Air (<code>brew install cloudflared</code>)	5 min
3	Authenticate <code>cloudflared</code> to Cloudflare account	5 min
4	Create tunnel in Cloudflare dashboard	10 min
5	Configure tunnel: <code>localhost:8768</code> → <code>dashboard.systack.net</code>	10 min
6	Run <code>cloudflared</code> as service (launchd or keep open)	10 min
7	Test on iOS Safari	5 min
8	Update dashboard docs with new URL	10 min
Total		~1 hour

Rollback Plan

- Keep Tailscale serve running in parallel
- If Cloudflare Tunnel fails, revert to `*.ts.net` URLs
- Zero downtime: both can run simultaneously

Fallback: Option 1 (Custom Domain A Record)

If Cloudflare Tunnel doesn't work: 1. Create subdomain `dashboard.systack.net` 2. Point A record to the public IP of the VPS (if public) or the Tailscale IP 3. Use `certbot` to get real LetsEncrypt certificate 4. Configure nginx or similar reverse proxy

This requires either: - Public IP for the MacBook Air (not typical), OR - A VPS in the middle that proxies to the MacBook via Tailscale

More complex but fully independent.

Immediate Workaround (While Implementing)

For current iOS users: 1. Open Tailscale iOS app 2. Connect to tailnet 3. Use Chrome or Firefox instead of Safari (they may handle the cert differently) 4. OR: Access dashboard via desktop and use the mobile responsive view there

Files to Update After Fix

FILE	UPDATE
<code>index.html</code>	Remove any hardcoded <code>.ts.net</code> references, update help text

FILE	UPDATE
SAOS-Dashboard-Mobile-Access-Guide-*.md	Add iOS cert trust section
systack-site/saos/	Update onboarding links if any point to <code>.ts.net</code>

Decision Required

Green — which option do you want?

OPTION	EFFORT	COST	UX
A. Cloudflare Tunnel	1 hour	\$0	✔ Perfect
B. Custom domain A record	2 hours	\$0	✔ Perfect
C. Manual iOS cert install	30 min	\$0	✘ Terrible
D. Do nothing (use desktop)	0	\$0	⚠ Limiting

My recommendation: **Option A (Cloudflare Tunnel)** — 1 hour, zero cost, solves the problem permanently.

Ready to implement when you confirm.